

Appendix M

Detailed Analysis of Expert Rating Explanations

M.1 Excellent

At the highest level, outputs rated as Excellent are characterised by a complete and coherent synthesis of the literature. These outputs demonstrate a comprehensive understanding of the subject matter, fully integrating all foundational themes and critical arguments from the source material.

Within this category, the *full pipeline* represents the strongest and most consistent performance. All five experts who assigned an Excellent rating, highlighted the system’s ability to integrate all foundational themes and major arguments present in the corpus. In terms of analytical depth, two experts explicitly noted the identification of research gaps, while one observed systematic comparison and contrast between studies. Structural coherence was also a defining feature: one Expert identified a clear and sustained line of argument both within and across paragraphs, and another emphasised the use of thematic clustering to organise the discussion. Overall, the defining strength of the *full pipeline* lies in its complete coverage

of the literature, as evidenced by the unanimous recognition of foundational themes in the output.

By contrast, Excellent performance was far less frequent in the *partial pipeline*. Experts identified only a single instance in which the output successfully integrated the foundational themes derived from the source papers. *ChatGPT* outputs were not associated with this category.

Taken together, these findings indicate that only the *full pipeline* consistently achieves the level of integration and analytical depth required for Excellent performance, while the *partial pipeline* does so only rarely and *ChatGPT* does not reach this level.

M.2 Good

Unlike the Excellent category, experts did not converge on a single, fixed definition of what constitutes a Good output, as interpretations varied depending on the tool. Nevertheless, a shared baseline emerged across all three systems: outputs in this category consistently moved beyond mere summary and demonstrated an ability to identify research gaps within the literature.

Strengths Observed Across Systems

For both the *full pipeline* and the *partial pipeline*, Good ratings were associated with a relatively high level of scholarly synthesis and structural coherence. In terms of coverage, both systems generally captured the foundational themes present in the source papers. Their analytical approach extended beyond simple description, as they were able to establish meaningful comparisons and contrasts between studies. This level of critical engagement was supported by a clear logical structure, typically organised through thematic clustering, which enabled a coherent progression of ideas across sections. In addition, experts noted that these outputs were useful for crosschecking purposes, as they occasionally facilitated

the identification of novel research directions. The academic language employed was also considered appropriate, with accurate use of domain-specific terminology.

Limitations Observed Across Systems

Despite these strengths, both the *partial pipeline* and *ChatGPT* exhibited notable deficiencies. With respect to coverage, outputs often lacked granular detail and displayed imbalanced citation practices. Their critical analysis was also more limited, frequently characterised by the presentation of research findings in isolation rather than through an integrated comparison across studies. As a result, the synthesis remained partial, and the broader relationships within the literature were not always made explicit.

Interpreting the Assignment of Good Ratings

To clarify the apparent inconsistency in how Good ratings were assigned to the *partial pipeline* and *ChatGPT*, a follow-up question was asked to the Experts. Specifically, they were asked to explain why they had assigned a Good rating when their accompanying explanation did not appear to be consistent with that evaluation. The experts' responses revealed that the value of the AI output is defined by its utility rather than its readiness for final use. For example, Expert 18 explained that although he would not use the result directly, it functioned as a "*valid first draft*" that supported the development of his own ideas. Similarly, Expert 10 highlighted that the "*organization of content*" offered immediate value, even though the text still required further "*polishing.*" In the same vein, Expert 21 justified the Good rating by describing the output as a "*work in progress.*"

Taken together, these accounts indicate that experts evaluate AI systems through a lens of pragmatic utility. Rather than expecting fully polished outputs, they prioritise the extent to which the tool reduces cognitive effort during the early stages of writing. At the same time, these findings also demonstrate that such evaluations are not lenient. On the contrary, experts engage in a rigorous and highly critical trade-off, recognising the usefulness of the output while remaining attentive to its limitations.

Therefore, a Good rating can be understood as indicating a high-utility prototype. At this level, the AI transitions from functioning solely as a text generator to acting as a preliminary scholarly collaborator. The defining characteristic of such outputs is their ability to capture the central themes of the source material and organise them into a coherent thematic structure that supports further development. Although the text typically requires refinement to meet publication standards, it provides immediate value.

Comparative Performance Across the Three Systems

Within this category, the *full pipeline* represents the strongest performance. Its outputs are distinguished by a consistently well-developed organisational structure, in which studies are logically grouped and synthesised with a relatively high degree of analytical depth. The resulting narrative demonstrates a level of coherence and intentionality that approximates that of a human-authored text, supported by accurate citation practices and a close alignment between headings and content.

The *partial pipeline* occupies an intermediate position, offering a useful structural foundation but with less consistency in analytical depth. Its primary strength lies in the creation of a clear organisational framework that can support further interpretation and idea generation. However, this is offset by limitations in coverage, particularly a lack of granular detail, and a tendency to present findings in isolation rather than as part of an integrated synthesis. Consequently, the output is valued more for its structural guidance than for its analytical completeness.

Finally, ChatGPT represents the baseline for a positive evaluation within this category. Its outputs provide functional value primarily at a structural and diagnostic level. While they are capable of identifying research gaps and offering an initial organisational framework, they frequently lack depth and exhibit uneven integration of evidence, particularly in the distribution of citations. As such, a Good rating in this case reflects recognition of the tool's role in reducing the effort required for initial drafting, even when the content remains rudimentary and requires substantial human intervention to reach academic standards.

Empirical Evidence of the Ratings

A more detailed examination of the empirical findings further illustrates the distinctions observed across the three tools. This analysis draws on expert evaluations to provide a granular account of how each system performs within the Good category, with particular attention to coverage, analytical depth, structural coherence, citation practices, and linguistic quality. By comparing these dimensions systematically, the following discussion highlights not only the relative strengths of each tool but also the specific limitations that shape their overall performance. This deeper level of analysis serves to substantiate the comparative patterns identified earlier, offering a more nuanced understanding of how and why the *full pipeline*, *partial pipeline*, and *ChatGPT* differ in their ability to support the development of a coherent and analytically grounded literature review section.

For the *full pipeline*, scholars consistently highlighted a sophisticated organisational blueprint, with studies logically clustered by shared themes in all five instances. Within this structure, a clear and explicit line of argument was maintained in three observations, ensuring coherent progression through appropriate transitions. The analytical capacity was also strong: four cases demonstrated effective comparison and contrast between studies, while three explicitly identified research gaps.

Coverage was similarly robust. Foundational themes and major arguments were accurately integrated in six out of nine cases. Structural precision further supported this performance, including two instances where headings accurately reflected the content and one instance of rigorous alignment between headings and the corresponding text. Citation practices were also positively evaluated, with two observations noting equitable distribution of references and two confirming that all citations were authentic and derived directly from the provided corpus. In one instance, the output facilitated the identification of a novel research idea previously unrecognised by the expert.

Linguistically, the *full pipeline* achieved a high level of scholarly quality. In two out of three observations, the narrative tone resembled that of a human-authored text, while

one instance was noted for its readability. This was reinforced by accurate use of scientific terminology in one case. Overall, the Good rating for the *full pipeline* is driven by its strong performance in coverage, logical structure, and critical analysis, resulting in outputs that are structurally robust and analytically credible.

The *partial pipeline*, by contrast, exhibited a more uneven performance. A lack of granular detail was the most frequent issue, identified in three out of five instances. Although foundational themes were included in one case, another observation noted that unnecessary detail detracted from the scholarly focus. Issues of sourcing fidelity also emerged, including one instance of secondary referencing that failed to attribute the original author. Citation layout presented additional limitations, with one observation noting concentration of references at paragraph endings. Despite this, the output demonstrated some utility, as one instance supported the identification of new research ideas.

Analytical depth showed similar inconsistencies. A failure to identify research gaps was reported in three out of seven cases, alongside two instances where findings were presented in isolation without synthesis. These weaknesses were partially offset by isolated successes, including one instance of gap identification and one of effective comparison between studies. The progression of ideas was also mixed: one observation highlighted a clear and logical line of argument, whereas another identified abrupt transitions between sections. Nevertheless, thematic clustering was recognised as a structural strength in one instance.

From a linguistic perspective, the *partial pipeline* demonstrated high terminological accuracy but limited surface-level polish. One observation highlighted correct conceptual definitions, while another noted accurate use of domain-specific terminology. However, these strengths were undermined by one instance each of mechanical and compositional errors. Overall, the *partial pipeline*'s weaknesses are concentrated in coverage, particularly insufficient detail and in critical analysis, especially the limited identification of research gaps. While it retains strengths in structural organisation and the identification of new research ideas, these are offset by isolated summarisation and technical flaws.

For *ChatGPT*, outputs rated as Good were characterised by uneven citation distribution, with two out of two observations noting that references were concentrated at the ends of paragraphs. In terms of content, the writing was generally acceptable but lacked sufficient depth in three out of four cases; in one instance, the omission of core themes was the primary issue. Analytical performance was divided: one case demonstrated successful identification of research gaps, while an equal number of observations reported a failure to compare findings, with studies presented in isolation.

The factors underpinning a Good rating for *ChatGPT* are primarily related to coverage and citation practices. The most frequent issue was insufficient detail, followed by uneven citation distribution. These findings indicate that, while the output provides a functional starting point, it lacks the depth and integrative sophistication required for scholarly use without substantial revision.

Summary of Good

Overall, the Good category reflects outputs that provide clear practical value as structured and analytically informed starting points, while still requiring refinement. The *full pipeline* demonstrates the strongest and most consistent performance, the *partial pipeline* offers a useful but less complete structural foundation, and *ChatGPT* provides baseline utility primarily in supporting early stage drafting.

M.3 Fair

In contrast to the Good category, a Fair rating reflects outputs that are functionally usable but exhibit substantial weaknesses requiring significant human intervention prior to scholarly use. At this level, the text demonstrates a recognisable attempt at academic synthesis. However, it is undermined by inconsistencies in analytical depth, structural coherence, and technical precision. The defining characteristic of this category is a lack of reliability. While the general research landscape may be captured, the connection between evidence and argu-

ment remains weak or incomplete. As a result, the generated content fails to maintain the rigorous standards of attribution, narrative continuity, and technical vocabulary expected in professional academic discourse.

Shared Limitations Across Systems

Across all three systems, a shared technical limitation emerges in the distribution of citations. In multiple instances, references are disproportionately clustered at the ends of paragraphs rather than being integrated throughout the prose. This pattern contributes to a fragmented reading experience and weakens the perceived linkage between claims and supporting evidence. In addition, both the *partial pipeline* and *ChatGPT* exhibit consistent deficiencies in coverage and narrative development. Experts frequently noted insufficient detail, resulting in underdeveloped arguments, alongside a lack of coherent progression, which prevented the text from forming a logically structured discussion. These issues were further compounded by inaccuracies in terminology, with some outputs relying on inappropriate or imprecise language that diminished their academic credibility.

Empirical Evidence of the Ratings

Despite being assigned a Fair rating, the *full pipeline* differs qualitatively from the other systems, as its limitations are primarily attributable to isolated errors rather than systemic weaknesses. The most prominent issue concerns citation accuracy. In one instance, a scholar identified a misattribution of authorship, initially interpreting it as a hallucination. However, upon reviewing the provenance of the claim, the expert revised this evaluation, recognising that the error originated from a misattribution present in the source material itself rather than from the system:

Ah, okay, then it makes sense. It's just that this paper made a mistake when citing Gao, and that's why the tool is reproducing that error. It's not a hallucination, but rather what it found in a reference. ... But of course, the person who wrote this was wrong, because the original article doesn't say that. So I don't

know exactly how to evaluate it, although it means it wasn't a hallucination as I thought; actually, the hallucination belonged to whoever wrote this (Expert 4).

This observation highlights an important distinction: while the *full pipeline* may reproduce inaccuracies, these are inherited from the corpus rather than generated independently. As with the issue of secondary referencing discussed in Section ??, the concern raised by Expert 4 arises from the tool's reliance on verifiability rather than strict factual accuracy. A further limitation identified was the occasional clustering of citations at paragraph endings. However, this phenomenon is more appropriately interpreted as a minor and discrete deviation within an otherwise structurally robust output.

The *partial pipeline*, by contrast, demonstrates more consistent and systemic shortcomings. The most significant issue identified by experts is a failure of logical flow, with all seven evaluators noting a complete absence of narrative progression. This resulted in disjointed sections lacking a coherent argumentative thread. Coverage related issues were also prominent, although less uniform in nature. While two experts highlighted insufficient detail, another observed that the inclusion of irrelevant information detracted from the overall scholarly focus. Citation practices presented mixed results: two scholars identified clustering of references at paragraph endings, whereas one considered the distribution to be appropriately balanced. Linguistic precision was also compromised, with one reviewer noting the persistent use of incorrect terminology, which weakened the authoritative tone of the text.

Taken together, the Fair rating for the *partial pipeline* is primarily driven by deficiencies in logical structure, particularly the absence of narrative flow, alongside recurring issues in coverage and citation distribution. Although some variability exists in specific observations, the overall pattern indicates a lack of internal coherence and analytical integration.

For *ChatGPT*, Fair rated outputs are characterised by more pronounced deficiencies across multiple dimensions. Coverage limitations were especially salient, with four out of eight instances indicating insufficient detail and limited analytical engagement. In addition, experts identified instances of content that were not grounded in the provided corpus, in-

cluding two cases where claims were entirely absent from the source papers. Issues of source validity were further reinforced by two observations of fabricated citations. Structural inconsistencies also emerged, including one case where headings did not align with the underlying content and another where headings were inappropriately named.

The analytical dimension of *ChatGPT*'s outputs further contributed to the Fair rating. In two out of four cases, the text relied on isolated summarisation of individual studies without establishing comparative or integrative links. This was accompanied by a failure to identify research gaps and, in one instance, an overreliance on a single author's findings to the exclusion of broader perspectives. Logical flow was also problematic: two observations highlighted the misuse of transitional markers and a lack of coherent progression between ideas. More broadly, the organisational structure was criticised in two cases for being disjointed and lacking a clear framework.

Finally, issues of academic language were particularly prominent. In three instances, experts identified the use of inaccurate or inappropriate scientific terminology, undermining the technical credibility of the text. Citation distribution also remained inconsistent, with one observation noting clustering at paragraph endings.

Overall, the Fair rating for *ChatGPT* is driven by compounded weaknesses across coverage, critical analysis, logical structure, and academic language. The outputs frequently fail to move beyond descriptive summarisation, lack coherent organisation, and exhibit significant issues in both source validity and terminological precision.

Summary of Fair

Overall, the Fair rating reflects outputs that demonstrate partial academic functionality but lack the reliability required for scholarly use. While the *full pipeline* is affected primarily by isolated errors inherited from source material, the *partial pipeline* exhibits systemic weaknesses in logical structure and coverage. *ChatGPT*, by contrast, shows compounded deficiencies across coverage, analysis, structure, and academic language, resulting in outputs that require substantial revision before they can be considered academically credible.

M.4 Poor

In contrast to the higher rating categories, a Poor rating reflects outputs that fail to meet fundamental standards of scholarly reliability and structural coherence. At this level, the generated text exhibits a breakdown across multiple core dimensions, including factual accuracy, citation integrity, and logical organisation. The defining characteristic of this category is a double failure of reliability. First, the output fails the veracity test, as it includes fabricated information or misattributes claims to incorrect authors. Second, it fails the structural test, producing disjointed and incoherent narratives that lack a clear logical progression. As a result, the text becomes difficult to follow and does not demonstrate the critical engagement expected in academic discourse.

Shared Limitations Across Systems

Across all three systems, several shared deficiencies emerge. A prominent and consistent issue is the fabrication of information, whereby claims or findings are introduced that are entirely absent from the provided set of papers. In parallel, all tools exhibit a failure to maintain cohesive narrative structure, frequently generating content that is disorganised and lacks a clear, logical sequence. This combination of factual inaccuracy and structural disintegration significantly undermines the credibility and usability of the outputs.

Despite these shared weaknesses, important differences exist in how each system manifests Poor performance. Both the *partial pipeline* and *ChatGPT* demonstrate a common tendency toward citation misattribution, where claims are incorrectly linked to authors or papers that did not originally present them. This indicates a systemic difficulty in maintaining accurate connections between evidence and source material, leading to fundamental violations of academic attribution standards.

Empirical Evidence of the Ratings

For the *full pipeline*, Poor ratings are primarily driven by failures in content boundaries and structural logic. In two instances, experts observed that the generated text included information entirely absent from the provided corpus, indicating a breakdown in maintaining

strict grounding in the source material. Structural weaknesses were also evident, with two observations highlighting a lack of clear organisational strategy and overall disorganisation. These issues were compounded by limitations in analytical depth; in one case, the output disproportionately relied on a single source or author, failing to incorporate a balanced range of perspectives. Additionally, narrative flow was compromised in one instance, where ideas moved abruptly between sections without clear rationale or explicit connections, resulting in a fragmented discussion of the literature. Taken together, these findings indicate that Poor performance in the *full pipeline* is primarily associated with deficiencies in coverage fidelity and macro-level organisation, rather than isolated technical errors.

The *partial pipeline* exhibits Poor performance through a combination of weaknesses in referencing accuracy and technical precision. Experts identified two instances of incorrect attribution, where claims were linked to the wrong authors or papers. In terms of coverage, one case involved the inclusion of information not present in the source material. Citation practices also revealed inconsistencies, including one instance of uneven distribution of references across the section. Structurally, the output demonstrated limited coherence, with one observation describing the organisation as disjointed and lacking a clear framework. Analytical depth was similarly constrained, as the text occasionally remained at a basic descriptive level, presenting findings in isolation without establishing meaningful connections or comparisons between studies. Issues in academic language further contributed to the Poor rating: in one instance, terminology was fundamentally incorrect, while in another, specialised terms were introduced without adequate definition or contextualisation. Overall, the *partial pipeline*'s performance at this level is characterised by persistent challenges in citation accuracy and the imprecise use of domain-specific language, which undermine both credibility and clarity.

For *ChatGPT*, Poor rated outputs are marked by pronounced deficiencies in source validity and logical coherence. Experts identified multiple instances of fabricated or invalid references, including two cases where cited works were entirely absent from the provided

corpus and one case of incorrect attribution of a claim to an unrelated author. Structural issues were also prominent, with two observations noting that the text moved between ideas without clear transitions or rationale, resulting in a lack of rhetorical cohesion. In terms of coverage, one instance included content not grounded in the source material, while another lacked sufficient detail for a scholarly review. Organisational weaknesses were similarly evident, with one evaluation describing the output as completely disorganised and lacking a recognisable structural strategy. Finally, stylistic limitations were noted, including one instance where the writing appeared overly mechanical and artificial. Taken together, these findings indicate that *ChatGPT*'s Poor performance is primarily driven by failures in citation integrity and logical flow, combined with weaknesses in coverage and stylistic delivery.

Summary of Poor

Overall, a Poor rating reflects a critical breakdown in the core functions required for academic writing. While all three systems demonstrate issues at this level, the underlying causes differ. The *full pipeline* is primarily affected by failures in maintaining strict adherence to source material and coherent organisation, whereas the *partial pipeline* struggles with attribution accuracy and technical language. *ChatGPT*, by contrast, exhibits the most pronounced issues in source validity and narrative cohesion. These distinctions highlight that, although all systems can produce unusable outputs under certain conditions, the nature of their limitations varies systematically across tools.

M.5 Very Poor

Unlike the higher rating categories, the Very Poor rating reflects outputs that are functionally unusable for scholarly purposes due to systemic failures across both structural organisation and content integrity. At this level, the generated text lacks a coherent macro-level organisational blueprint, resulting in a presentation of research that is scattered and disjointed rather

than systematically developed. The literature is neither logically sequenced nor grouped according to a meaningful framework, creating significant barriers to comprehension.

In addition to these structural deficiencies, outputs in this category exhibit fundamental breakdowns at the micro-level. The narrative frequently shifts between ideas without clear rationale, and transitional markers are often misused, creating the appearance of logical continuity where none exists. More critically, Very Poor outputs fail to maintain grounding in the provided source material. This includes the introduction of fabricated claims, the use of references not contained within the predefined corpus, and misalignment between section headings and their corresponding content. As a result, such outputs fail to meet even the minimal standards of academic writing and require complete reconstruction rather than incremental revision.

Shared Limitations Across Systems

Across the two evaluated systems, the defining characteristic of the Very Poor category is the combination of failures in both coverage and logical structure. Experts consistently emphasised the absence of a coherent organisational strategy, alongside the inclusion of content that lacked fidelity to the source material. These issues collectively undermine both the credibility and usability of the generated text.

Empirical Evidence of the Ratings

For the *partial pipeline*, Very Poor performance was associated primarily with critical failures in logical flow and organisational coherence. Experts identified one instance in which transition markers were used in a manner that did not align with the reasoning of the sentences they connected, resulting in a breakdown of argumentative clarity. In addition, one case was characterised by a complete absence of a recognisable organisational blueprint, with the generated text described as entirely disorganised. These findings indicate that the primary weaknesses of the partial pipeline at this level lie in its inability to sustain both rhetorical coherence and structural integrity.

ChatGPT outputs rated as Very Poor exhibited more extensive and compounded deficiencies across multiple dimensions. In terms of coverage, experts identified two instances in which the generated content included claims and details entirely absent from the provided source set. In a further case, the output demonstrated a complete lack of alignment between sectional headings and the content they were intended to structure. Issues of source validity were also evident, including one instance in which references were introduced that did not originate from the predefined corpus.

Structural deficiencies were particularly pronounced. In two cases, the output was described as entirely disorganised and lacking any coherent organisational strategy. Logical flow was similarly compromised, with one instance highlighting abrupt transitions between ideas within sections without clear justification or linkage. Linguistic execution also contributed to the Very Poor rating, as one observation noted the use of specialised terminology without adequate introduction or definition.

Taken together, these findings indicate that Very Poor performance in *ChatGPT* is driven by pervasive failures in both coverage fidelity and organisational coherence. The most frequent issues include the inclusion of fabricated or non-grounded content, the absence of a clear structural framework, and the breakdown of logical progression. These deficiencies, combined with inaccuracies in citation practices and weaknesses in academic language, result in outputs that do not meet basic scholarly standards and cannot be meaningfully revised without substantial reconstruction.

Summary of Very Poor

Overall, a Very Poor rating reflects a complete breakdown in the fundamental requirements of scholarly writing, rendering outputs functionally unusable. While both systems exhibit severe deficiencies at this level, the underlying causes differ in scope and intensity. The *partial pipeline* is affected primarily by failures in logical flow and the absence of a coherent organisational framework. *ChatGPT*, by contrast, demonstrates more pervasive breakdowns across coverage fidelity, structural coherence, and citation validity, including the introduc-

tion of non-grounded information and misalignment between structure and content. These distinctions indicate that, although both systems fail to meet minimal academic standards in this category, ChatGPT exhibits a broader and more compounded set of limitations.